

DataMind AI

AI-Powered Excel & CSV Data Cleaning System

Submitted To	Miss Nida
Course	Artificial Intelligence
Submitted By	Aqib Ali(23226), Ubaid(23238), Ahmad(23243)
Department	Information Technology
University	Govt. MAO College, Lahore
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1. Project Overview

DataMind AI is an intelligent data cleaning and analysis system that leverages Google Gemini AI to help users clean, process, and transform raw Excel and CSV files. Unlike conventional data cleaning tools that blindly apply transformations, DataMind AI first analyzes the uploaded file, identifies data quality issues, and then interactively asks the user targeted questions before making any changes. This ensures that every cleaning decision is intentional and user-approved.

The core innovation of this project lies in its human-in-the-loop AI pipeline: the system does not process data automatically. Instead, it acts as an intelligent assistant that understands the data context, generates relevant questions in checkbox and radio-button form, and then executes cleaning operations strictly based on the user's preferences. The final cleaned file can be downloaded in the user's preferred format — either Excel (.xlsx) or CSV.

2. Problem Statement

Data professionals, businesses, and students regularly deal with raw, unstructured datasets that suffer from a variety of quality issues. These include:

- Duplicate records caused by repeated data entry

- Missing or null values in critical columns
- Inconsistent date formats across different rows (e.g., DD/MM/YYYY vs YYYY-MM-DD)
- Inconsistent text casing (e.g., 'LAHORE', 'lahore', 'Lahore' for the same city)
- Outliers and anomalous numeric values that skew analysis
- Incorrectly typed data stored in wrong column formats

Existing tools either require programming knowledge (pandas, R) or apply transformations blindly without user control. DataMind AI bridges this gap by providing an accessible, AI-guided interface that is suitable for non-technical users while remaining powerful enough for data professionals.

3. Project Objectives

1. Develop a web-based system that accepts Excel (.xlsx, .xls) and CSV file uploads.
2. Implement an AI analysis engine using pandas to detect data quality issues automatically.
3. Integrate Google Gemini AI to generate context-aware, data-specific cleaning questions.
4. Present questions to users in an intuitive checkbox and radio-button interface.
5. Execute cleaning operations based exclusively on user-provided answers.
6. Generate an AI-written natural language summary of all transformations applied.
7. Allow users to download the cleaned file in their preferred format (CSV or Excel).

4. Proposed Solution & Architecture

The system follows a three-stage pipeline:

4.1 Upload & Analysis Stage

The user uploads a raw Excel or CSV file through the React frontend. The FastAPI backend reads the file using pandas and performs a comprehensive analysis: detecting column data types, counting missing values per column, identifying duplicate rows, flagging outliers in numeric columns, and detecting inconsistent formatting in text and date columns. This analysis is passed to the Gemini AI model.

4.2 AI Question Generation Stage

The Gemini AI model receives the file analysis and generates 4–8 targeted, context-specific questions. Each question is tied to an actual detected issue. For example, if the 'Salary' column has 3 missing values, Gemini generates a question specifically about that column with actionable

options such as 'Fill with mean', 'Fill with 0', 'Drop rows', or 'Keep as-is'. Questions are returned as structured JSON and rendered as checkboxes or radio buttons in the frontend.

4.3 Processing & Export Stage

After the user answers all questions and selects their preferred output format, the backend executes the cleaning operations in sequence using pandas. A summary of all changes is generated by Gemini AI in plain English. The cleaned file is saved and made available for download in the user's chosen format.

5. Technology Stack

Frontend	React.js (Vite) with custom CSS — Upload, Questions, and Results pages
Backend	FastAPI (Python) — REST API for upload, question generation, and processing
AI Model	Google Gemini 2.5 Flash API — Question generation and summary writing
Data Processing	Pandas — File parsing, analysis, and cleaning operations
File Export	openpyxl (Excel), pandas (CSV) — Cleaned file generation and download
Environment	Python-dotenv — Secure API key management via .env file
State Management	React Context API — Global app state across multi-step workflow

6. Key Features

6.1 Multi-Format File Support

Users can upload files in .xlsx, .xls, or .csv format. The system auto-detects the file type and uses the appropriate pandas parser. This makes the tool universally applicable across different data sources.

6.2 Intelligent Question Generation

Rather than applying a fixed set of rules, DataMind AI uses Gemini to read the actual file analysis and generate questions that are relevant to the specific dataset. Two different files will produce two completely different sets of questions, making the tool adaptive and intelligent.

6.3 Human-in-the-Loop Processing

No data is modified without explicit user approval. Every cleaning operation — whether removing duplicates, filling missing values, or handling outliers — is executed based on the user's answers to AI-generated questions. This ensures transparency and user control.

6.4 Flexible Output Format

At the end of the workflow, users are presented with a choice of output format: Excel (.xlsx) or CSV (.csv). The cleaned file is then generated in the selected format and made available for immediate download.

6.5 AI-Generated Summary

After processing, Gemini generates a short plain-English summary describing exactly what was changed in the dataset — how many duplicates were removed, which columns had missing values filled, and what the final row count is. This acts as an audit trail for the cleaning process.

7. Expected Outcomes

Upon successful completion, DataMind AI will deliver:

- A fully functional web application accessible via browser with no installation required
- An AI pipeline capable of analyzing Excel and CSV files and generating relevant cleaning questions
- A processing engine that applies user-directed transformations using pandas
- A downloadable cleaned file in CSV or Excel format with an AI-generated change summary
- Clean, documented source code with a comprehensive README for setup and usage

9. Conclusion

DataMind AI presents a novel approach to data cleaning by combining the analytical power of pandas with the intelligence of Google Gemini AI and the accessibility of a modern web interface. The project addresses a real-world problem faced by data users at all skill levels and delivers a solution that is interactive, transparent, and user-controlled.

The system's unique human-in-the-loop design ensures that AI acts as a guide rather than an autonomous agent, giving users full confidence in the transformations applied to their data. This project will demonstrate practical application of AI APIs, file processing pipelines, and full-stack web development — key competencies in modern software engineering.

Submitted by: Aqib Ali | 23226